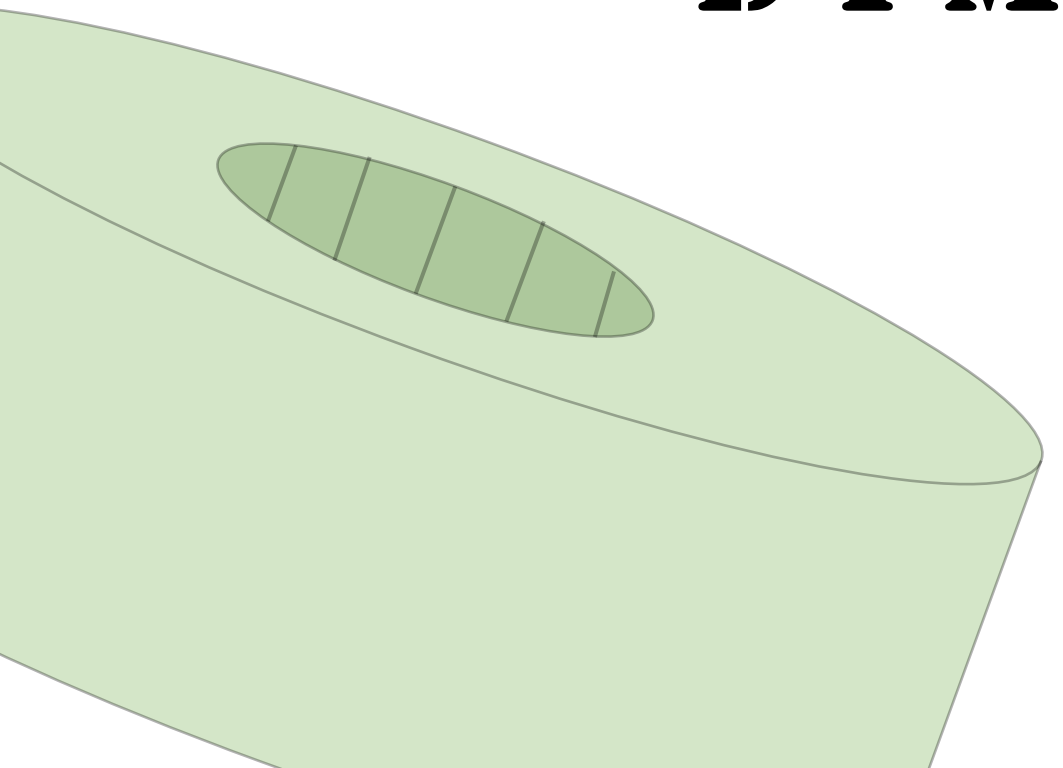
DIMENSION

강병찬, 주하늘



Introduction - 'Dimension'?

- Intuitively, we determine the dimensions of various spaces.
- But what means by the term ‘dimension’?
- Number of parameters? Cardinality?

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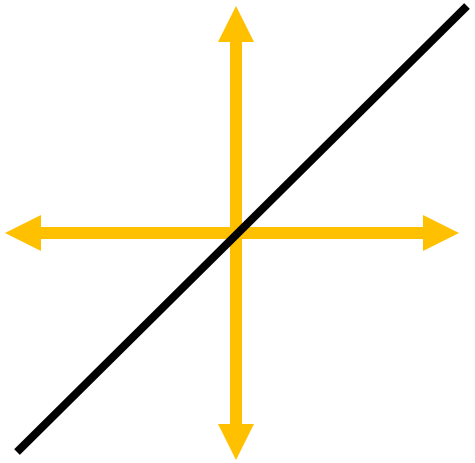

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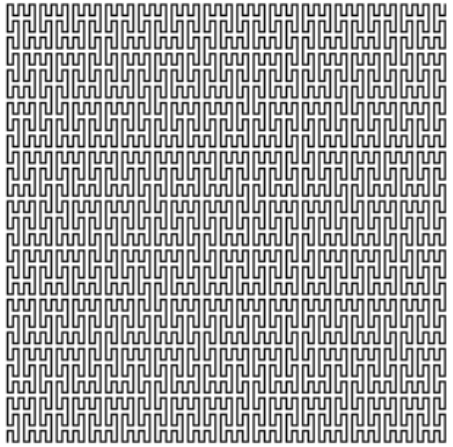

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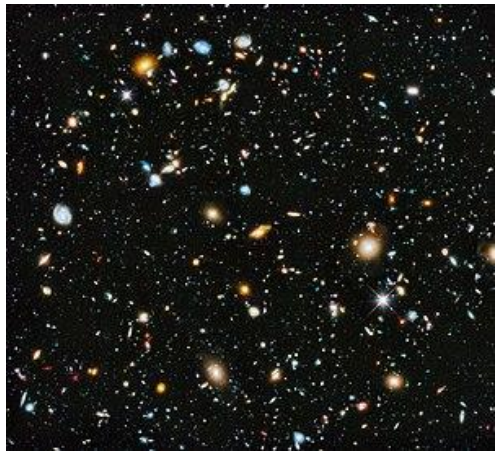

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Contents

- Prelude
- Definition of dimension
- Application : Dimension of $\mathbb{R}^n$
- Imbedding; Imbedding theorems
- Hausdorff Dimension

Prelude - Order of Subset Collection

$(X, \mathcal{T})$: Topological space

$\mathcal{A} \subseteq \mathcal{P}(X)$: Subset collection

Order: maximal natural m s.t

$$(1)\ \exists A_1, \dots, A_m \left(\bigcap_{k=1}^{m} A_k \neq \emptyset\right)$$

$$(2)\ m < n \Rightarrow \bigcap_{k=1}^{n} B_k = \emptyset \quad (\forall B_1, \dots, B_n \in \mathcal{A})$$

Prelude – Refinement (1)

$\mathcal{A}, \mathcal{B} \subseteq \mathcal{P}(X)$: Subset collections / covers

Refinement: $\mathcal{B}$ s. t.

$$\forall B \in \mathcal{B} \quad \exists A \in \mathcal{A} \text{ s.t. } B \subseteq A$$

Prelude – Refinement (2)

(Ex)

$$S^1 = \{(x, y) : x^2 + y^2 = 1\}$$

$$L_a \stackrel{\text{def}}{=} \bigcup_{\theta \in [0, 2\pi)} \{\{(\cos(x+\theta), \sin(x+\theta)) : x \in [0, a]\}\} \quad (0 < a < 2\pi)$$

- L_a: cover of S^1, $a_1 < a_2 \Rightarrow L_{a_1}$ refines L_{a_2}

Q. Then, the order of L_a is ⋯

Prelude - Homeomorphism

$(X, \mathcal{T}_X)$, $(Y, \mathcal{T}_X)$: Topological spaces

- Homeomorphism: bijection $f: X \to Y$ s.t.

  (1) $f^{-1}(V) \in \mathcal{T}_X$ whenever $V \in \mathcal{T}_Y$

  (2) $f(U) \in \mathcal{T}_Y$ for all $U \in \mathcal{T}_X$

- $X \cong Y \Leftrightarrow \exists$ homeomorphism $f: X \to Y$

Q. $(-1,1) \cong \mathbb{R}$ (T / F)

Prelude - Imbedding

$(X, \mathcal{T}_X)$, $(Y, \mathcal{T}_X)$: Topological spaces

- Imbedding:injection $f: X \to Y$ s.t.

(1) $f^{-1}(V) \in \mathcal{T}_X$ whenever $V \in \mathcal{T}_Y$

(2) $X \cong f(X)$

Prelude - Manifold

$(X, \mathcal{T})$: Topological space

- m−manifold :

$$\forall x \in X \ \exists U \in \mathcal{T} \ \exists V \subseteq \mathbb{R}^m \ x \in U \cong V \ (V\text{: open}).$$

(Ex) $\Delta = \{(x, x) : x \in \mathbb{R}\} \subseteq \mathbb{R}^2$: 1−manifold.

Prelude - Smooth Structure

$(X, \mathcal{T}_X)$: m-manifold

- Chart: (U, ϕ) s.t. $U \in \mathcal{T}_X$, $U \cong^{\phi} V$ where $V \subseteq \mathbb{R}^m$ open.
- Atlas $\mathfrak{U}$: $\{(U_\alpha, \phi_\alpha) : \alpha \in J\}$ where $\bigcup_{\alpha \in J} U_\alpha = X$.
- Transition $\tau_{\alpha,\beta}$: $\phi_\alpha(U_\alpha \cap U_\beta) \to \phi_\beta(U_\alpha \cap U_\beta)$ defined as $\phi_\beta \circ \phi_\alpha^{-1}$.

  (Remark. $U_\alpha \cap U_\beta$ should be nonempty to define transition.)

- Smooth structure: each $\tau_{\alpha,\beta}$ is smooth.

Dimension

$(X, \mathcal{T})$: Topological space

$\mathcal{A} \subseteq \mathcal{T}$: Open cover

Dimension: smallest natural m s.t.

$\forall \mathcal{A}$, $\exists \mathcal{B}$: open refinement of $\mathcal{A}$ s.t $ord(\mathcal{B}) \leq m + 1$

$(X, \mathcal{T})$: Topological space

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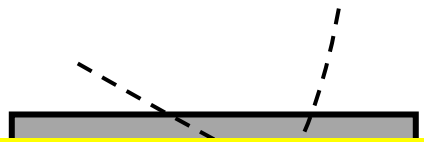
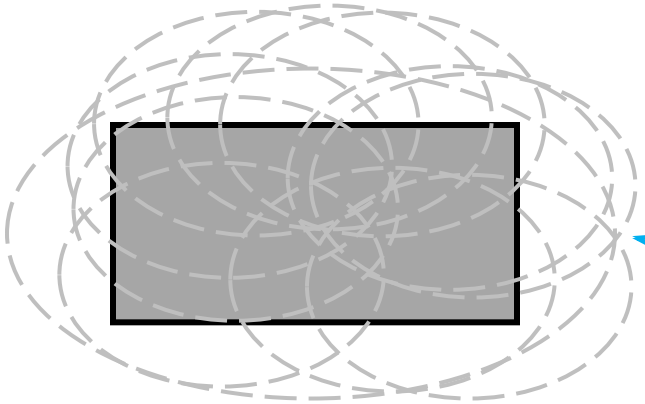
Dimension



1. Find the minimal order of every refined open cover of $\mathcal{A}$.

2. Take the maximum over every $\mathcal{A}$.

- Why not just the order of $\mathcal{A}$?



$\mathcal{A}$: 응애 나 오픈 커버

$\dim(X) = 100$ ^^^^77

Application : $\mathbb{R}^n$

Our intuition says:

$\langle$All compact subset of $\mathbb{R}^n$ has dimension at most n.$\rangle$

Q. Can we prove this?

Application : $\mathbb{R}^n$

spoiler alert : at the end, we are not achieving the complete proof.

Divide the proof into two parts: Let $X \subseteq \mathbb{R}^n$ be compact.

- $\forall d > 0$, $\exists \mathcal{B}$: open cover of X s. t.

$$\forall \mathrm{B} \in \mathcal{B}(\mathrm{diam\ B} \leq d) \text{ and } ord\mathcal{B} \leq n + 1.$$

- $\mathcal{A}$: given open cover of X. Then $\exists d > 0$ s. t.

$$\forall U \in \mathcal{B}(\mathrm{diam}(U) < d) \Rightarrow \forall U \in \mathcal{B}(\exists A \in \mathcal{A} \quad U \subseteq A) \qquad (\mathcal{B}\text{: open cover})$$

Application : $\mathbb{R}^n$

- 다음의 예시를 생각해 보자.

➢ 벡터공간 $\mathbb{R}^n$의 basis $\mathrm{B} = \{v_1,\ v_2,\ \cdots,\ v_n\}$:

각 벡터의 크기– δ, 임의의 두 벡터의 내적 – $\frac{\delta^2}{2}$(임의의 두 벡터의 사잇각이 $\frac{\pi}{3}$).

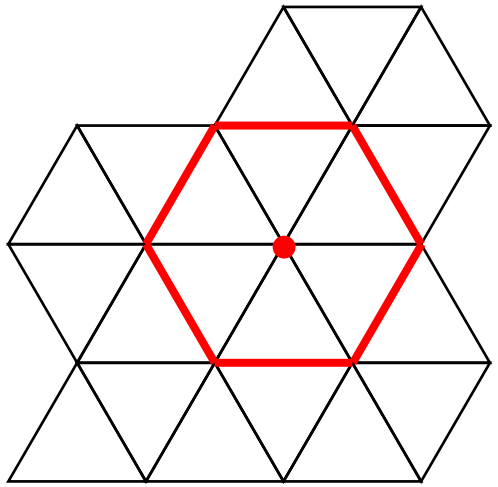
Ex) 2차원에서 벡터들은 한 정삼각형의 이웃한 두 변을 이루며, 3차원에서 세 벡터는 한 정사면체의 한 꼭짓점에 incident한 세 변을 이루게 된다.

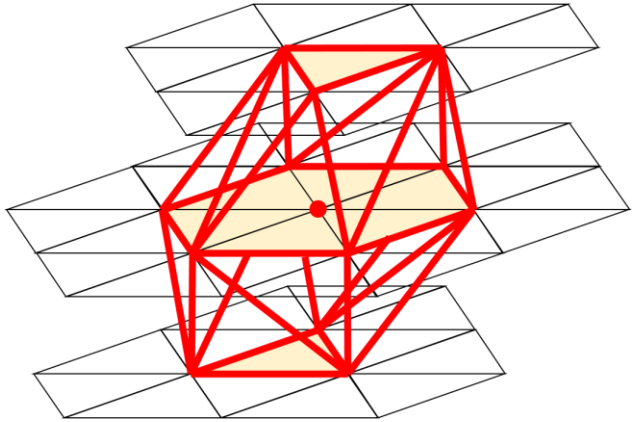
➢ B의 벡터들의 '정수계수 선형결합' ($\sum c_i v_i$, c_i는 정수)으로 표현되는 벡터를 위치벡터로 가지는 점들의 집합 : S.

Application : $\mathbb{R}^n$

격자점들 사이의 최소 거리는 δ이다.
거리가 δ인 두 점이 서로 adjacent하다.

- 이렇게 잡은 S는 마치 삼각형 모양이 반복되는 격자의 격자점들의 집합이 된다.
- 그림 : 각각 $\mathbb{R}$, $\mathbb{R}^2$, $\mathbb{R}^3$ 위의 격자. 격자의 한 점과 adjacent한 점들을 찾고 그들 중 adjacent한 것끼리 서로 잇기 → $\mathbb{R}^2$에서는 정육각형, $\mathbb{R}^3$에서는 '육팔면체'.
- $\mathbb{R}^n$에서 유사하게 경계를 잡고, 경계와 그 내부를 포함하는 도형을 고려한다.





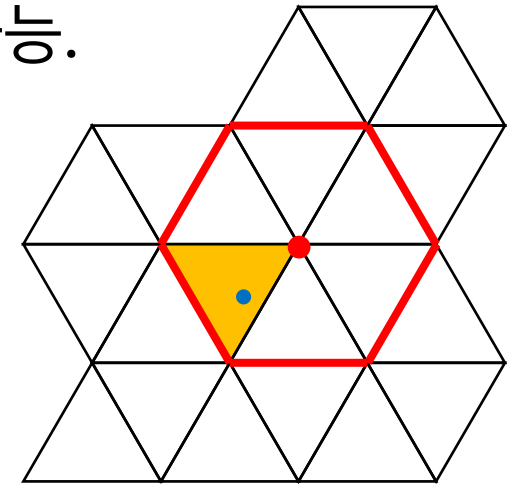
이 그림은 잘못되었음!!!

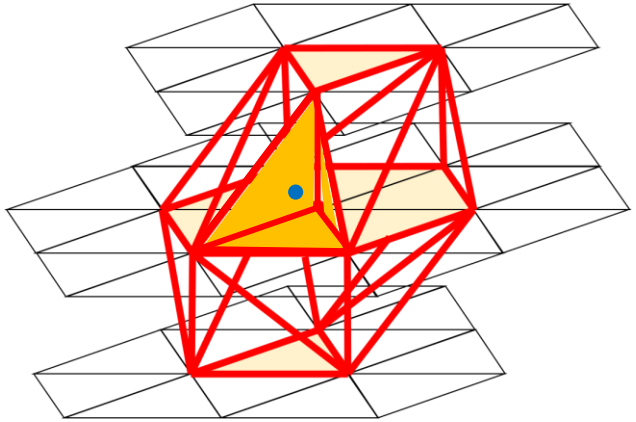
Application : $\mathbb{R}^n$

임의의 d에 대해, $\mathbb{R}^n$ 의 compact subset X의 open cover $\mathcal{B}$가 존재하여 $\mathcal{B}$의 모든 원소의 직경이 d 이하이고 $ord\mathcal{B} \leq n+1$이다.

- 해당 도형의 interior를 격자점에 대응하는 open set으로 두고, **모든 격자점에 대해** 그러한 open set을 잡은 뒤 X를 덮음으로써 open cover $\mathcal{B}_\delta$를 만든다.
- 각 open set의 직경 : 2δ. (δ는 임의로 설정 가능.)
- (엄밀X)그림에서 1, 2, 3차원 simplex 안에 놓인 파란 점들을 보면, simplex의 꼭짓점들에 대응되는 open set들**만**이 파란 점을 포함.

- n차원에서 이러한 확장이 가능.
- $ord\mathcal{B}_\delta \leq n+1$.



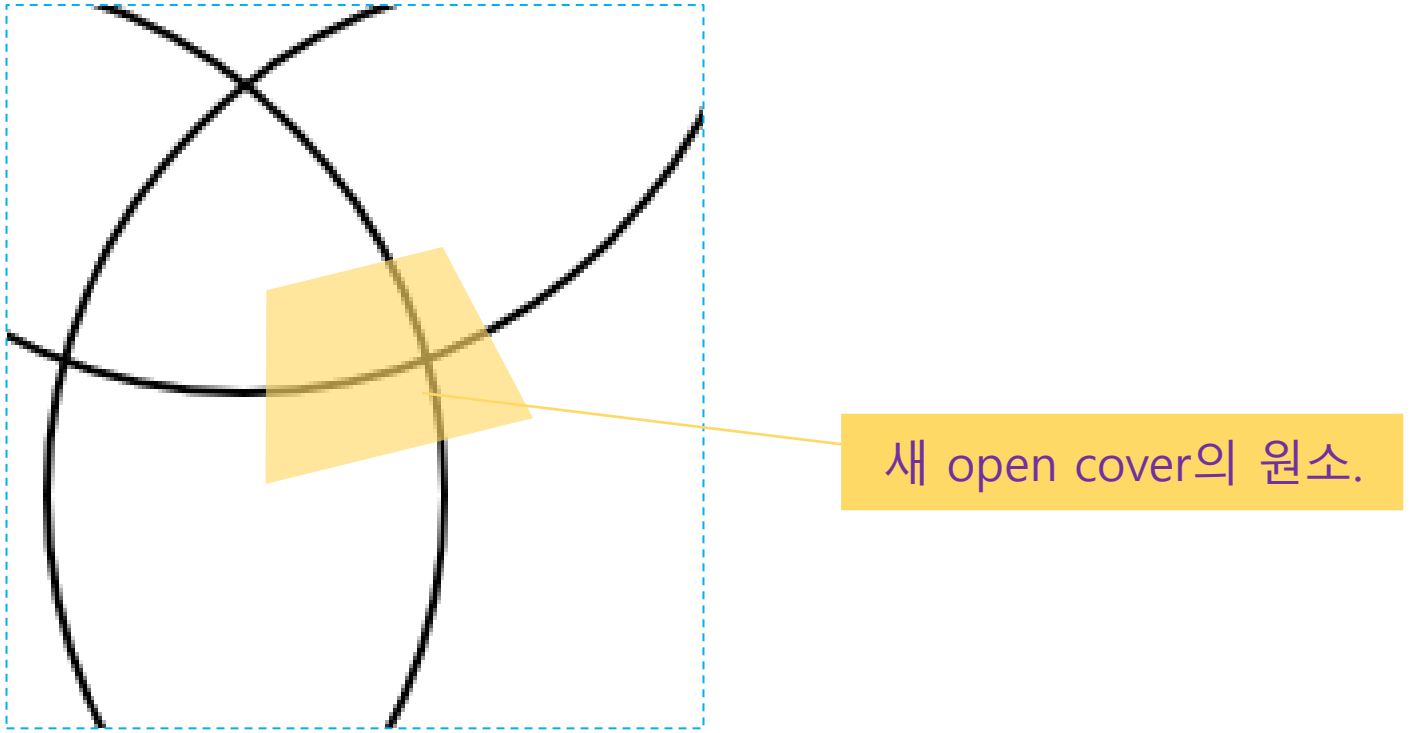


잘못 그렸는데 다시 그리기 귀찮았어요

Application : $\mathbb{R}^n$

- Let $X \subseteq \mathbb{R}^n$: compact. If $\mathcal{A}$: open cover of X, then $\exists d > 0$ s.t

$$\forall U: open, \operatorname{diam}(U) < d, \qquad \exists A \in \mathcal{A} \quad U \subseteq A$$



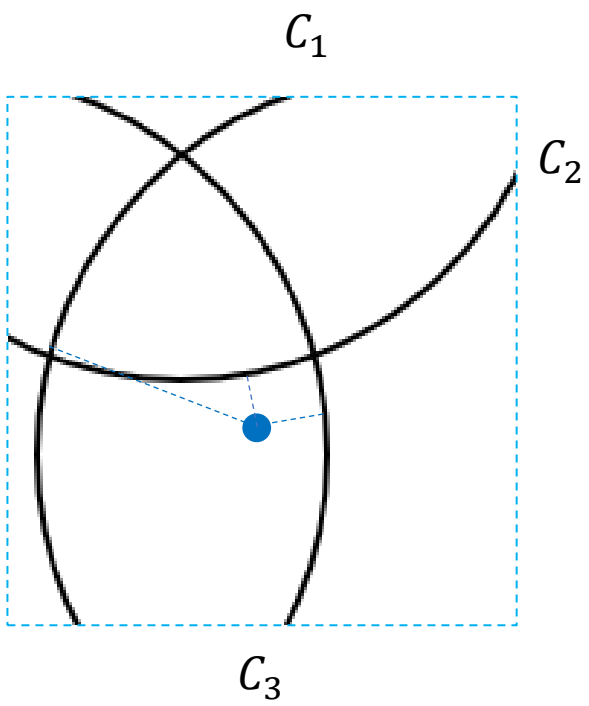
Application : $\mathbb{R}^n$

For $\mathscr{A} = \{C_1, C_2, \cdots, C_n\}$, let us define $\delta = \inf_{x \in X} \frac{1}{n} \sum_{i=1}^{n} \inf_{a \notin C_i} |x - a|$.

For any x, there is some k s.t.

$$\frac{1}{n} \sum_{i=1}^{n} \inf_{a \notin C_i} |x - a| \leq \inf_{a \notin C_k} |x - a|.$$

$\therefore U$: open, $x \in U$, $\mathrm{diam}(U) < \delta \Rightarrow U \subseteq C_k$.



Application : $\mathbb{R}^n$

$$\delta = \inf_{x \in X} \frac{1}{n} \sum_{i=1}^{n} \inf_{a \notin C_i} |x - a|.$$

- Trouble: $\delta > 0$?
  - It is known that $\delta > 0$ (Lebesgue's Number Lemma).
  - Munkres proved the theorem for $\mathbb{R}^n$ in different way (from that I did).

Q. For compact $X \subseteq \mathbb{R}$, $\dim(X) = 1$ if and only if ⋯

Generalization - to Manifolds

Our intuition also says:

$\langle$Nonempty m-manifold has dimension at most m.$\rangle$

- Invariance of Domain says:

$$m \neq n \Rightarrow m\text{-manifolds} \neq n\text{-manifolds}$$

$$\Rightarrow m\text{-manifold} : m\text{-dimensional!}$$

Another Question: "Similar Space" of m-Manifolds in $\mathbb{R}^n$

- Homeomorphic $\approx$ "Similar"

Q. Is there a subset $Y \subseteq \mathbb{R}^n$ homeomorphic to given m−manifold X?

Imbedding theorem

- (Imbedding theorem) Every m-dimensional compact metrizable space X can be imbedded into $\mathbb{R}^{2m+1}$.

- (Corollary) Every compact m-manifold can be imbedded into $\mathbb{R}^{2m+1}$.

  ($2m+1$ is the minimal natural that suffices it)

- (Corollary) Compact metrizable space is finite-dimensional $\Leftrightarrow$

  it can be imbedded in some finite-dimensional $\mathbb{R}^n$.

Strong Whitney imbedding theorem

(Strong Whitney imbedding theorem)

- Any smooth real m-dimensional manifold, which is Hausdorff and Second-countable, can be smoothly imbedded in $\mathbb{R}^{2m}$.

Smooth Imbedding(function)?

$(X, \mathcal{T}_X, \mathfrak{U}_X), (Y, \mathcal{T}_Y, \mathfrak{U}_Y)$: finite-dimensional smooth manifold

- Smooth function: $f: X \to Y$ such that

1. $\forall x \in X$, $\exists (U, \phi) \in \mathfrak{U}_X$ s.t. $x \in U$.
2. $\exists (V, \psi) \in \mathfrak{U}_Y$ s.t. $f(x) \in f(U) \subseteq V \in \mathcal{T}_Y$
3. $\psi \circ f \circ \phi^{-1}: \phi(U) \to \psi(V)$ is smooth.

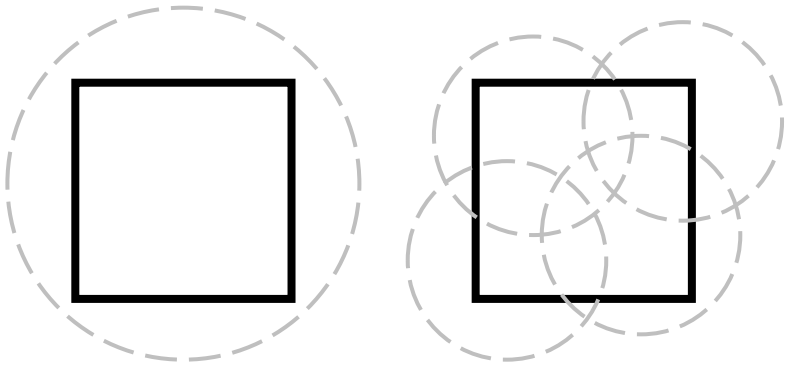
Hausdorff Dimension

- Extension of dimension from naturals to reals (in geometry)
- Precise definition :

$$\dim_H(X) := \inf\{d \geq 0 \,:\, \mathcal{H}^d(X) = 0\}, \text{ where } \mathcal{H}^d(S) = \lim_{r \to 0} \inf_{\forall C} \left\{ \sum_i r_i^d \right\},$$

where C can be any cover of S by balls with radii $0 < r_i < r$.

- Ex) When $r_1 = r_2 = \cdots = r_n = R$, $\sum_i r_i^d = nR^d$.
- We find that $n \propto R^{-(dimension)}$ (see figure):
- So that $\mathcal{H}^d(S)$ equals 0 iff $d > (dimension)$.



Hausdorff Dimension

- In fractals in Euclidean spaces(Say $\mathbb{R}^n$), Hausdorff dimension can be calculated by the expression

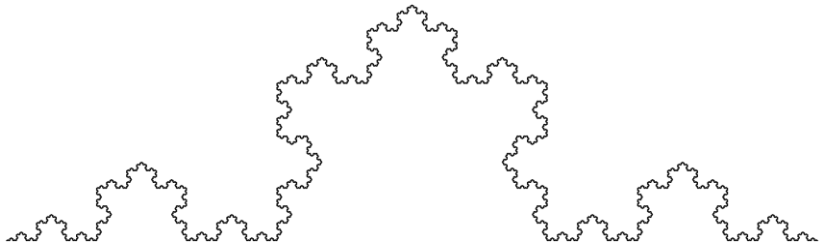
$$\log_{(length\ ratio)}(n - dimensional\ \ 'amount'\ ratio)\,.$$

- That is; “What is the expansion ratio of the length/area/volume of the fractal if I expand the fractal with a given length ratio?”

Honorable mention : Minkowski Dimension(Similar to Hausdorff’s : ‘1’ for $\mathbb{Q}$.)

Hausdorff Dimension

0.6309



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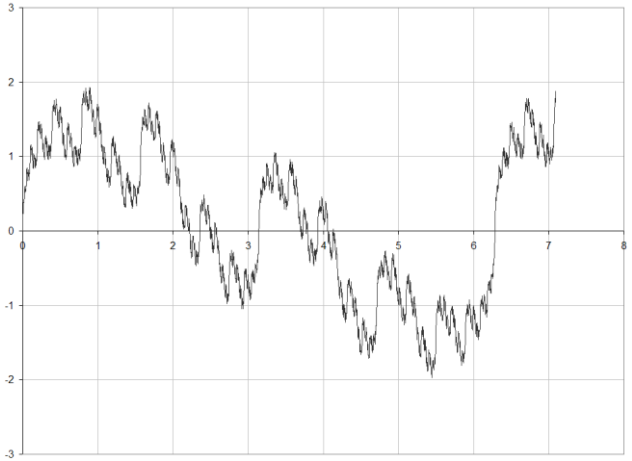

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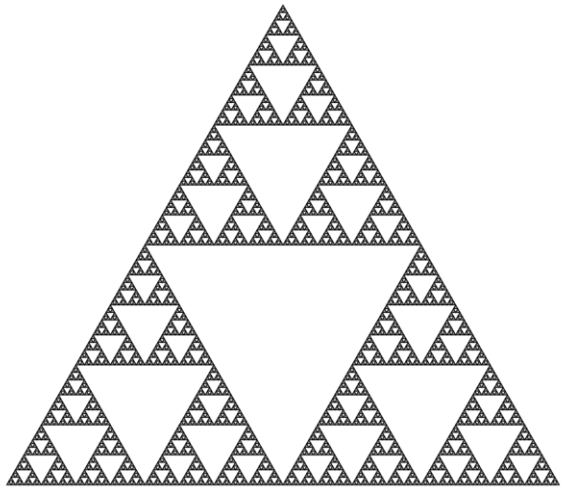

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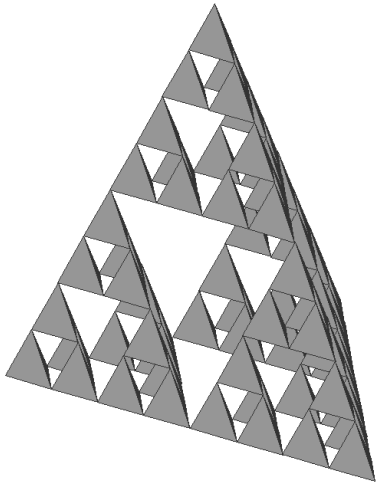
Hausdorff Dimension



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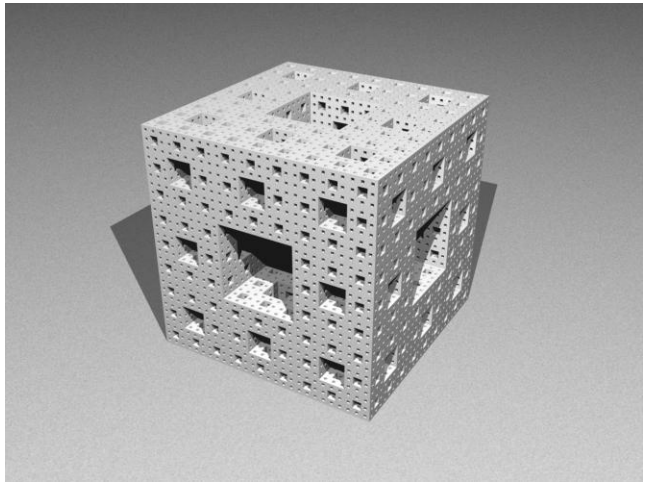
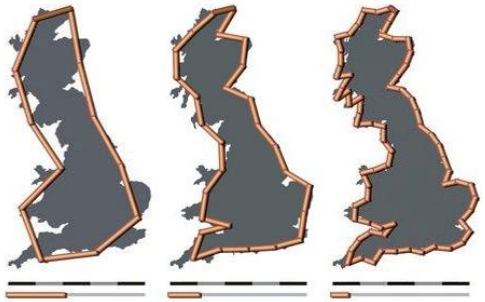


2.7268

Coastline of
Great Britain



1.25

Coastline of
Norway

1.52

Finale